

While We Wait. . . How Users Perceive Waiting Times and Generation Cues during AI Image Generation

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Abstract

While AI text generation is instantaneous, image generation and reasoning models take longer. To signal processing time to users, platforms tend to use interface cues such as progress bars, skeleton screeners, and textual descriptions of the generation process. However, little is known about whether and how these cues affect user experience of generative AI tools. Through semi-structured interviews with 11 participants who interacted with various AI image generation tools, we examined user perceptions of waiting times and progress indicators, which we term “generation cues.” Data show that users are generally accepting of, sometimes even value, waiting times, viewing them as an inherent part of the creative process. However, they rarely notice generation cues. These findings challenge traditional usability principles of speed, efficiency, and feedback, suggesting the need to rethink waiting times and generation cues in the context of generative AI. We offer design recommendations for leveraging waiting times and generation cues to enhance human-AI collaboration in creative domains.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Image generation; Waiting time perception; Generation cues; Human-AI collaboration; User experience design

ACM Reference Format:

Hui Min Lee, Davis Yadav, Sangwook Lee, Keerthana Govindarazan, Cheng Chen, and S. Shyam Sundar. 2025. While We Wait. . . How Users Perceive Waiting Times and Generation Cues during AI Image Generation. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719725>

1 Introduction

While tools such as Midjourney, Stable Diffusion, and Dall-E can significantly increase creative potential, they also bring with them longer processing times compared to text generating AI tools [4]. Unlike almost instantaneous text generation or web browsing at high WiFi speeds, AI image generation can take anywhere from several seconds to a few minutes. Recognizing these increased waiting times, platforms have implemented visual progress indicators, such as progress bars and skeleton screeners, to signal to users that the system is processing their prompt. However, these indicators may take on new meaning in the context of generative AI, potentially affecting users’ perception of the AI generation process. For example, recently introduced advanced reasoning models, such as OpenAI’s o1 and Deepseek’s R1, reveal their step-by-step reasoning process, providing a sense of transparency that can enhance users’ perceptions of waiting periods and provide insight into AI’s generation process.

Through semi-structured interviews with participants using different AI image generation tools, this study examines how users perceive and make meaning of the image generation process. We focus on the psychological experience of waiting for AI-generated creative output, as well as the role of progress indicators (henceforth referred to as generation cues) on users’ perceptions of AI-generated images. Our findings show that users are generally accepting of, and even value, waiting times, viewing them as an inherent part of the creative process. Users associated longer waiting times with better

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719725>

quality outputs, revealing an implicit mental model regarding processing time. However, they rarely noticed generation cues. These insights are crucial for both researchers and platform designers, as waiting times and generation cues may have important implications for users' attitudes towards AI, and can be thoughtfully designed to enhance user experience and human-AI creative collaboration.

2 Related Work

2.1 Image Generation Waiting Times

Waiting time is significantly higher when a Generative AI tool is creating images rather than text [4]. As users wait for their images to be generated, they may turn to existing mental models of how generative AI works, interpreting the waiting times in different ways. These interpretations can not only reveal folk theories—simplified explanations of new technologies—that users hold about how generative AI works [21], but also provide insight into why users engage with different AI image generation tools. For example, in a user experience study with DALL-E 2, Kim et al. [10] found that users often developed their own hypotheses to make sense of AI image generation, such as believing that “instead of capturing the exact meaning of a sentence, AI combines various elements in it” (p. 10) or “When AI represents an element it is not familiar with in its existing data set, it synthesizes elements that do not match exactly but are related” (p. 10). Notably, with respect to waiting times, individuals may also mindlessly apply various social rules of human-human interaction to AI image generation, such as interpreting quicker generation times as the AI being more skilled or efficient, in line with the central proposition of the Computers as Social Actors paradigm (CASA) [19]. Conversely, Buçinca et al. [2] suggest that waiting times can be a cognitive forcing function that increases thoughtful elaboration of an AI's generated output, while Joshi and Vogel [9] found that adding a time delay before submitting a prompt encouraged more effortful prompting.

There are a variety of other mental models that users may apply, such as perceiving that longer waiting times mean that the AI is working harder like a human would be [11], or that spending a longer time to create a creative output means that the generated output will be more valuable, much like their beliefs in other creative contexts. This may be explained by a buildup of anticipation, in line with the concept of delayed gratification [17]. Users' perceptions of waiting times can also be strongly influenced by the expectations that they have as a result of prior interactions with the tool [10]. These expectations go beyond perceived waiting times of how long it takes to generate an image, but may also involve their expectations about the quality of image that the AI will generate given the time taken to generate an image [10]. In sum, waiting times may hold a variety of important psychological meanings for users, but have not fully been explored in the context of AI image generation. This current study thus aims to confirm and extend these insights in the specific context of waiting times during AI image generation.

2.2 Generation Cues in the AI Context

To counteract potential negative effects of waiting time, interface designers have historically relied on progress indicators that inform users about the status of a task in progress and the remaining time

required. Visual progress indicators in traditional interfaces such as websites and mobile applications help users gauge progress, reduce anxiety and improve wait-time satisfaction [24]. Indeed, Hohenstein et al. [8] found that interactive animations decreased perceived waiting time and increased user satisfaction compared to progress bars or passive animations. Another study found that perceived waiting time was shortened when users were presented with a decelerating progress bar [15].

Different progress indicators can elicit different psychological responses. For example, skeleton screens are visual placeholders mimicking the page layout to reassure users during brief load times (under 10 seconds), preventing assumptions of error in the loading of the page [30]. In line with the Nielsen Norman Group's recommendations for interaction design [20], delays of longer than 10 seconds can cause shifts in attention and focus, while such progress indicators can emphasize the visibility of the system's status, showing how the system operates independently of user control. Research further indicates that animated progress bars, particularly those with percent-done indicators, significantly improve user satisfaction and decrease perceived wait times compared to “spinners” or static text [1, 33]. These cues can even be combined to provide users with more information. For instance, adding clarity to spinners or progress indicators by including explanatory text, such as “Loading comments...” can enhance user experience [24].

Today, these progress indicators are commonly adopted and leveraged by generative AI platforms as well. Because of the longer waiting time, AI image generation tools frequently include visual progress indicators. For instance, MidJourney, a leading AI platform for image generation, uses skeleton screens consisting of four square screens to show images as they are being created. Similarly, DALL-E, integrated into ChatGPT, uses a screener to display the remaining time for image creation. A common feature across platforms such as Bing Image Creator is a progress bar that informs users of the time left to complete an image (Figure 1).

In this study, we regard these progress indicators as *generation cues*, which we consider a catchall term for visual signals indicating to users that the AI generation process is ongoing. We choose to term these “generation cues” because although they are traditional visual progress indicators, they also go beyond that to involve specific textual descriptions of the AI's generation process, such as “Creating an image with Imagen,” seen in the case of Google Gemini. This broader definition allows us to examine the full spectrum of indicators used during AI image generation.

2.3 Effects of Generation Cues in AI Image Generation

Various psychological outcomes are possible depending on the type of generation cues. In some image generation platforms, animated visuals show the step-by-step progress of the generation process, making the experience both interesting and engaging for users [8]. For example, in Starry AI, users first see a clock interface with spinning hands, indicating that the system is processing, and then see a paintbrush indicator swiping up and down on a skeleton screener, which indicates that image generation is underway. Liao et al. [16] discuss the importance of interface design to reveal the “black box” of generative AI. Animation styles, as seen in the case of Starry AI,

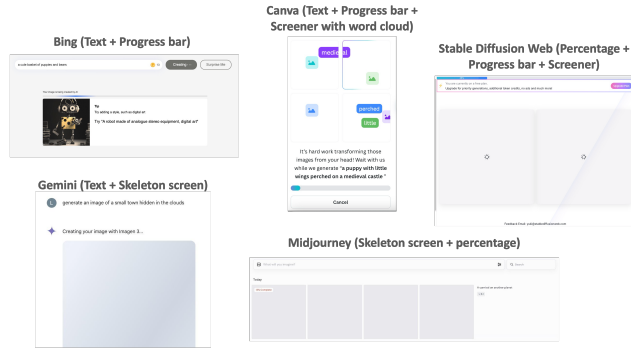


Figure 1: Generation Cues in Various Image Generation Tools

may allow users to understand each stage of the generation process in a visually appealing way that enhances enjoyment. Generation cues can be an avenue to serve as a transparency cue, also activating various cognitive heuristics (or mental shortcuts), such as the “helper heuristic” (the perception that the AI is helping users, hence trustworthy) or the “control heuristic” (the perception that the user has control over the actions and consequences of AI, therefore it’s a credible system) [12, 27, 28].

Furthermore, phrases such as “It’s hard work creating an image!”, which appears as a Canva generation cue, or “Creating your image” seen in Google Gemini, can anthropomorphize the AI, highlighting creativity as a characteristic of machine agency, as opposed to more traditional machine functions such as data processing or retrieval of information. This implicitly ascribes to GenAI the notion of “mind perception” [6], which has two components—agency, the ability to think, plan and act; and experience, the ability to feel emotions and consciousness—that are typically considered uniquely human. For instance, some individuals hold the belief that because AI does not have intelligence, thoughts, or emotions, it cannot ever be credited as a creator of art [7]. Across two experimental studies, Messingschlager and Appel [18] found that AI was ascribed with lower agency and experience than a human artist, and thus reduced appreciation for the generated art.

On the other end of the spectrum, certain generation cues that signal more machine-like functionality, using phrases such as “Your image is being created by AI,” may signal greater machine agency, potentially diminishing users’ sense of ownership or contribution to the generated image [28]. Generation cues may also imply a certain level of automation, which has been shown in previous studies to put users in conditions of mental underload and result in less satisfactory user performance [32].

Hence, this study extends previous work on users’ perceptions of AI image generation tools [10] and the current understanding of traditional progress indicators [24], seeking to understand how users interpret both waiting times and different generation cues in the context of generative AI. We pose the following research questions:

RQ1: How do users interpret the waiting times of AI image generation, and what are the psychological effects of perceived waiting time on users’ understanding of the AI image generation

process, their user experience, and perceptions of generated image quality?

RQ2: How do users interpret generation cues during AI image generation, and what are the psychological effects of generation cues on users’ understanding of the AI image generation process, their user experience, and perceptions of generated image quality?

RQ3: Is there any relationship between waiting time and types of generation cue, and do they affect perceptions of generated image quality?

3 Interviews

We conducted online interviews with 11 participants from December 2024 to January 2025. They were recruited via Prolific, and interviews were conducted on Zoom. The participant pool was diverse in terms of age, gender, and occupation (Table A1).

Each interview followed a semi-structured interview protocol comprising of three main stages. First, participants were asked about their prior experience with AI text and image generation tools, including their usage patterns and understanding of the technology. Next, participants were tasked to engage in a hands-on think-aloud protocol [5] by generating an image using three randomly selected AI image generation tools out of a pool of five (Midjourney, Stable Diffusion Online, Canva, Google Gemini, Bing Image Creator). We chose to provide participants with three tools to broaden the scope of our findings and ensure that participants’ responses were not based solely on their experience with a specific tool. We gave participants freedom to choose the image they wanted to create, occasionally suggesting topic ideas to them. We encouraged them to use the same prompt across all three tools to avoid variability of waiting times as a result of varying complexities of prompts. Finally, we asked about their emotional and cognitive responses to the tools, their perception of generation cues and waiting times, and their evaluation of quality of the image generated by the tool.

We transcribed the data and analyzed them using an open-coding technique, a process of “breaking down, examining, comparing, conceptualizing, and categorizing data.” [25]. Five of the authors coded the data independently, after which they discussed similarities and discrepancies in their respective findings.

4 Results

In this section, we present major themes that emerged from our analysis of participant interviews. We begin with users’ interpretations of waiting times, followed by the psychological effects of waiting, and finally, we examine the relationship between waiting times and generation cues.

4.1 Perception and Interpretation of Waiting Times (RQ1)

Participants showed a natural acceptance of waiting times, viewing them as an inherent part of the creative process. This acceptance manifested through two key mechanisms: quality expectations and creative effort attribution.

4.1.1 Output-Time Association. Participants consistently associated longer waiting times with better quality outputs, revealing an

implicit mental model about the relationship between processing time and output quality. As P10 explained: *“I’d have to wait a little bit longer for it to be accurate. That’s completely fine with me.”* This association was often rationalized through familiar life experiences, as P6 reflected: *“That’s sort of a basis for a lot of other things in our lives, right? So if you wait longer, the end result’s supposed to be better.”* P11 also applied the common cognitive heuristic of longer time meaning more output, and noted that *“I think because it’s (Gemini) only creating one image, so that’s why I feel like it was faster [than others].”* Such findings not only reveal the types of heuristics that users apply to AI, but may also reveal how users justify the “sweet spot” of constraint in creative tasks [22].

Indeed, this acceptance had its limits. When waiting times exceeded users’ expectations, they expressed dissatisfaction, indicating a violation of their current heuristics about AI’s capabilities: *“It’s a little bit slower than I thought it would be”* (P5). This may imply the invocation of the machine heuristic [28], suggesting that users approach AI with specific stereotypes and expectations of how machines often work and their processing speeds; failure to fully meet these expectations seems to negatively impact their experience. In fact, this finding has been supported in previous studies, which found similar drop-offs in user satisfaction when website loading times exceeded user expectations [23]. According to Onarheim and Biskjaer [22], users can tolerate only a certain level of constraint, in this case a delay, before becoming dissatisfied.

4.1.2 Effort Signaling. A second key mechanism for accepting the wait was the attribution of creative effort to the AI. Participants often anthropomorphized the AI during the waiting period, interpreting longer waiting time as an indicator of deeper engagement with the creative task, much like a human. Examples of these are how P11 suggested that longer waiting times meant the AI was *“work[ing] harder,”* while P5 interpreted it as the AI *“thinking about deeper subjects.”*

4.2 Psychological Effects of Waiting Times (RQ1)

4.2.1 Mental Models of AI Processing. Several mental models emerged as users tried to understand how AI processes their requests during the waiting period. Some participants anchored their understanding to human cognition: *“I assumed it was thinking in the way a computer thinks like, I don’t know how it works, but I assumed it was making like kind of synapses”* (P9). Others viewed it as a search process: *“[it is] gaining data and trying to search for results within the short period for the prompt you provided”* (P1), while some thought of it as an analyzer and creator: *“I think that the AI was analyzing what I typed into it, and basically like creating generating content based on my prompt”* (P2).

Interestingly, due to the uncertainty around how it actually functions, several participants found AI to be mysterious *“AI is kind of a mystery of how it works, what it does”* (P7), *“It’s kind of mysterious, but I’m assuming there’s some sort of mechanics behind it”* (P3). This sense of mystery came with an appreciation for its capabilities, particularly its rapid response time, as evident in P4’s statement *“I can only state to you that it is magical, like I said earlier I don’t know, because the time it’s taking is not as if there are people like working behind it. Within a few seconds, it displays the output I want. So, I*

really just cannot imagine how it is working. I just cannot.” These reflections reveal how limited understanding of AI and the waiting time for output generation can lead to both anthropomorphism and attribution of mystical qualities.

4.2.2 Disengagement with the Generation Process. Next, the speed of the generation affected how much participants engaged with generation cues during the generation process. Because the generation process appeared extremely quick, some users mentally disengaged during generation or engaged in other activities. For example, P9 said, *“My brain goes into elevator wait time, because I think my brain knows I don’t have time to get anything else done, so my brain is just like, okay, pause,”* while P8 finds ways to stay productive: *“I’ll pick up my phone and I’ll go...I may just go to Google and say, What does Google give me for that image? And then what does the AI give me for that image? So I’ll still be working on the project.”* As can be seen, waiting times also had the potential to turn users’ attention away from generation cues, which has implications for design. This may also explain why some users did not notice any of the generation cues.

4.3 User Engagement with Generation Cues (RQ2)

4.3.1 Limited Engagement with Generation Cues. While participants interacted with different image generation tools, it was interesting to see that out of 11 participants, seven of them did not actively notice or process the generation cues during the think aloud task unless explicitly asked to, as they were more focused on anticipating the output. For instance, P10 stated that *“I was so excited to see what the pictures were going to be. I wasn’t paying attention to everything else.”* As exemplified in P10’s comment, for some participants this anticipation manifested as excitement, while others experienced nervousness about their output and what kind of image would be generated *“I felt nervous at first about whether my prompts would work well”* (P11). These responses reflect the “gulf of envisioning,” an idea conceptualized by Subramonyam et al. [26] to convey the lack of certainty one feels when imagining the possible outcomes of LLMs.

In addition to anticipation, the quick response time of the tools (i.e., the speed at which the output was generated) made it difficult for participants to fully process the cues. As noted by P1, *“[The progress bar] was very fast before we got the results. So yeah, I didn’t really pay attention to it.”* Therefore, the emotional engagement in anticipation of output and fast response time of the systems narrowed participants’ attention and engagement with the generation cues, resulting in most of them failing to mention the generation cue when waiting for image generation, unless explicitly prompted to reflect on them.

4.3.2 Functional vs Creative Signaling of Generation Cues. Upon probing, most participants reported that they noticed some of the generation cues and recalled different cues like the spinners, loading bar etc., but merely as an *“afterthought”* (P11). Some participants reported that the cues provided an indication that their prompt had been registered and the tool was working: *“at least I know it’s working”* (P11). This falls in line with the intended function of progress indicators [24]. Overall, to most participants, there

were not really specific cues that stood out. However, for those who did comment on specific cues, their responses correspond to existing literature on the effects of different progress indicators [1, 33], where certain cues may serve informative signals, such as the fact that the system is working.

More specifically, progress bars with percentages were the most noticeable and more preferred. For example, P6 said, *“I like the progress. I’d say it’s more obvious how close it is to being complete.”* Since the progress bar was more animated and occupied more visual space, it sustained users’ attention, as noted by P6 *“Since there is a more defined beginning and an end, loading from the left to the right, and instead of it being in a smaller circle ... I feel like I’m more engaged with it, because you can actually see more of it.”* Additionally, percentages provided more useful information as they were psychologically more meaningful in indicating status of the task being performed by the system and acted as a feedback regarding their action on the system. For example, P5 stated that *“People like numbers. I like numbers. I’d like to see it go up, and when it reaches a certain point that means I know it’s done.”*

In addition to signaling AI’s status, P8 found the textual cues enlightening about the AI’s generation process. Referring to the word cloud in Canva (Figure 1), P8 pointed out that it signaled that *“you’re showing me exactly what it is that you’re drawing from. You’re using the words that I put in. So I know that you’re working on this and not something wrong that I may have accidentally typed.”* This assurance appears to have provided P8 with a sense of control or agency over the generation process, a key heuristic in usability that is shown to improve user experience [20, 28].

4.4 Relationship Between Generation Cues and Waiting Times (RQ3)

Beyond mental models and perceptions of the generation process, cues also impacted participants’ perception of waiting time. While they generally felt that the tools were fast in generating the output, their perception of waiting time varied and were a result of which tool they engaged with first or to what they were comparing, and the generation cues. For example, P11 pointed out the percentage indicators in Midjourney, commenting that *“the numbers made it slower... made it feel like it’s loading and loading and loading, which makes it feel like it’s longer when you got numbers.”* On the contrary, P2 perceived that the progress bars helped reduce his perception of waiting time, especially when images were generated before the progress bar loaded fully: *“it hasn’t hit a 100 percent yet, but we’ve got something generated here.”*

4.5 Attribution of Creative Responsibility

Finally, although not a part of what we had originally set out to study, an interesting finding that emerged was how users attributed creative responsibility for the generated image. A few participants commented that to them, creation was a collaborative effort: *“I feel like it’s about a 50-50 because it was my idea... I couldn’t take credit for generating these images”* (P10). Others attributed more agency to the AI: *“I think, the tool is probably most responsible for it... I’m kind of like directing it, telling it what to do, but it’s the one doing all”* (P3).

This attribution of creative responsibility was linked to users’ expectations of the AI and their process of waiting and prompting. Notably, when outputs did not meet users’ expectations, participants often blamed themselves rather than the AI: *“Maybe my expectations should be tempered a little bit”* (P9), *“I’m not good at putting prompts together”* (P11). This self-critical attribution suggests that waiting times may create investment in the outcome, causing users to take more personal responsibility for results.

5 Discussion

While conventional usability guidelines emphasize minimizing wait times for search engines and websites [13–15], our study found that users generally accepted and even valued waiting times in AI image generation, viewing the wait as an inherent part of the creative process. This acceptance seems to stem from two key mechanisms: (1) users’ social response to generative AI, thereby applying human creativity process expectations to image generation, and (2) individual mental models that associate longer waiting times with more positive outcomes.

These mechanisms suggest that users’ acceptance of the wait may, in fact, be a form of reconciliatory behavior to deal with the cognitive dissonance between their desire for immediate responses and the reality of the delays in AI’s image generation. Rather than expecting images to be generated quickly, users may be adjusting their expectations and acceptance thresholds during the waiting period. They rationalize their waiting times as necessary for obtaining quality output, thorough processing, or a greater number of images. Notably, unlike traditional satisficing situations where users lowered their expectations due to constraints, participants in our study sometimes elevated their expectations due to the wait, indicating an interesting shift in our understanding of waiting time from being a potential friction point to being a form of effort signaling.

Hence, unlike in traditional digital interfaces where waiting is seen as a source of friction, our findings suggest the potential benefits of having deliberate waiting times in generative AI contexts. This aligns with Cox et al.’s [3] concept of design frictions, where points of difficulty during interaction can enhance mindful engagement and prompt reflection, much like how P8 uses the waiting time to reflect on her own prompt and desired image. This is also reflected in a study by Joshi and Vogel [9], where intentional delays before prompt submission encouraged more effortful prompt creation.

Several theoretical frameworks help explain this phenomenon. The CASA paradigm [19] can explain why users mindlessly apply human creative process expectations to AI generation times. Just as people tend to appreciate artwork more when an artist spends more time creating it, participants attributed greater value to AI-generated images with longer generation times, mindlessly applying a social rule that is true for human endeavors. This same principle may explain users’ tendency to self-blame when generation results do not meet their expectations, as it mirrors human communication patterns where individuals often initially blame themselves for misunderstanding in a communication setting. In fact, this may also be due to the operation of machine heuristic [31], predisposing users to perceive AI to be accurate and free from error, thereby attributing blame to themselves. The waiting period

may also serve as a form of delayed gratification [29], potentially increasing users' appreciation and sense of ownership over the generated images. This suggests that, contrary to traditional usability goals of instant gratification, some form of waiting in generative AI contexts might enhance the overall user experience and output valuation.

Interestingly, while generation cues are implemented across all platforms, most participants reported minimal attention to these cues. This reduced awareness can be attributed to either strong anticipation during image generation or the quick processing times of current platforms. When noticed, users also expressed preference for progress bars and percent-done indicators, aligning with Nielsen's [20] principle of system status visibility. However, our findings suggest that traditional progress indicators may not fully capture the unique nature of AI generation processes at the moment, and more can be done to leverage the potential of generation cues for a better and more meaningful user experience.

Our findings also hold particular relevance given the emergence of reasoning models such as OpenAI's o1 and Deepseek's R1, which explicitly reveal their step-by-step thinking processes during generation. These descriptions of model reasoning constitute of form of generation cues, and an interesting question is the extent to which users may notice these as compared to traditional progress indicators. Such reasoning descriptions directly address the "black box" perception of current AI generation tools. By witnessing the AI's reasoning steps—how it interprets users' prompts, selects visual styles, and generate new output—users might develop more accurate expectations about outputs. Reasoning-based generation cues may also resolve the tension between limited engagement with traditional cues and users' genuine interest in understanding AI processes, potentially enhancing both user experience and creative collaboration. Ultimately, reasoning models provide room to productively fill users' waiting time with meaningful information about the system's thinking process, rather than merely signaling a delay. In the next section, we further highlight some key design implications for how cues can be implemented during waiting times.

5.1 Design Recommendations

Ultimately, traditional progress indicators aim to minimize perceived waiting times, but in creative AI contexts, such cues may take on new meaning. Our findings suggest several opportunities for rethinking generation cues to better align with user expectations and the creative nature of AI image generation. Designers may need to consider how to create meaningful waiting experiences, such as ways to signal the AI's creative effort, rather than simply to mark time passing or signal that the system is reacting to the user. This means going beyond traditional usability recommendations [20]. For example, rather than pursuing instant gratification, generation cues could be designed to provide insight into the AI's generation process. Canva's word cloud, which selects keywords from users' original prompts, may be a good exemplar in this regard. Such visualizations reveal the AI's internal processes, making wait time more cognitively engaging and allowing users to interpret the AI's generation process, rather than merely a delay to be tolerated. Future designs can thus more explicitly and creatively show how

the system interprets and processes user prompts, such as clearly showing the AI's reasoning steps.

Generation cues may also serve as a window into AI's "black box," addressing users' desire to understand these systems. To this end, we propose several specific design directions: first, current textual indicators often go unnoticed in favor of visual ones, possibly due to cluttered interfaces where several cues are present at once. Designers should consider more prominent placement of cues that signal the generation process, confirms the interpretation of the users' prompts, or provides context about system decisions. Interfaces could also be more minimalistic to direct attention to a specific generation cue, making them more effective as information channels rather than mere progress signals. To further improve perceived transparency of how the AI works, generation cues could also focus on communicating broader contextual information, such as the computational resources being used or environmental impacts of image generation, helping users understand the broader implications of AI use. Table A2 shows a more detailed summary of our recommendations.

5.2 Limitations and Future Directions

Our small sample is a limitation. Although sufficient for an exploratory qualitative study, it may not capture the full range of user experiences and interpretations. Additionally, our participants were interviewed about different AI image generation platforms, making it challenging to isolate the effects of specific generation cues. Future research can employ controlled experiments that examine how different generation cues affect user experience, allowing us to validate our findings with statistical support. Studies using eye-tracking or psychophysiological measures can also be employed to better understand attention to cues and/or assess user engagement as they are waiting.

Acknowledgments

This research was supported in part by the MSIT (Ministry of Science, ICT), Republic of Korea, under the Global Scholars Invitation Program (RS-2024-00459638) administered by IITP (Institute for Information & Communication Technology Planning & Evaluation).

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6 Appendices

Table A1: Participant Demographics and AI Usage

P#	Gender	Age	Occupation	Text Use	Image Use
P1	M	29	Asset manager	Occasionally	Never
P2	M	27	Data entry	Occasionally	Never
P3	F	23	Student	Great deal	Almost never
P4	M	33	Engineer	Great deal	Moderate
P5	M	18	Student	Great deal	Moderate
P6	M	41	Self-employed	Great deal	Never
P7	F	49	None	Occasionally	Never
P8	F	50	Project manager	Moderate	Moderate
P9	M	38	Lawyer	Occasionally	Occasionally
P10	F	54	Contractor	Never	Almost never
P11	M	33	None	Great deal	Great deal

Table A2: Design Recommendations and Implementation Strategies

Design Recommendations	Rationale	How to Implement
Meaningful Waiting Times to Improve User Experience	Users value waiting times as indicators of creative effort	Frame waiting as part of creative process: • Show image forming in stages (e.g., rough sketch to detailed rendering) • Show reasoning steps (e.g., "Interpreting prompt > Generating concepts > Refining details...") • "Effort signals" that highlight computational intensity (e.g., display CPU/GPU metrics)
System Transparency for Socially Responsible AI	Users express interest in understanding AI processes	Informative cues about how the generation process works: • Word cloud of prompt interpretation • Indicate type of training datasets that influence the current generation (e.g., "Drawing from artistic styles: impressionism") • Step-by-step visualization of generation process
Interface Simplification	Current cues often go unnoticed due to visual complexity	Minimalist design during generation: • Dimming non-essential UI elements • Generation cue on center of the page
User-Machine Balance of Agency	Users value sense of creative partnership	Acknowledge collaborative nature: • Highlight user's contribution to process (e.g., "Creating image with your idea") • Showcase AI's interpretation of input • Emphasize collaborativeness of outcomes generated
Error Prevention	Help users avoid common generation issues	Proactive guidance and error prevention: • Suggestions for generation parameters • Pre-generation checks with users to clarify prompts • Show slight variations of user's prompts that may yield better results
Cross-Platform Consistency	Help users build transferable mental models of AI generation	Standardized core elements: • Common generation phase terminology (e.g., "analyzing your prompt" or "refining your image") • Similar progress indicators patterns • Consistent generation cues • Shared prompt structure guidelines